



Stress-induced changes in body surface temperature are repeatable, but do not differ between urban and rural birds

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Abstract

Urbanisation can alter local microclimates, thus creating new thermal challenges for resident species. However, urban environments also present residents with frequent, novel stressors (e.g., noise, human interaction) which may demand investment in costly, self-preserving responses (e.g., the fight-or-flight response). One way that urban residents might cope with this combination of demands is by using regional heterothermy to reduce costs of thermoregulation during the stress response. In this study, we used black-capped chickadees ($n_{\text{urban}}=9$; $n_{\text{rural}}=10$) to test whether known heterothermic responses to stress exposure (here, at the bare skin around the eye): (1) varied consistently among individuals (i.e., were repeatable), and (2) were most pronounced among urban individuals compared with rural individuals. Further, to gather evidence for selection on stress-induced heterothermic responses in urban settings, we tested: (3) whether repeatability of this response was lower among birds sampled from urban environments compared with those sampled from rural environments. For the first time, we show that heterothermic responses to stress exposures (i.e. changes in body surface temperature) were highly repeatable across chronic time periods ($R=0.58$) but not acute time periods ($R=0.13$). However, we also show that these responses did not differ between urban and rural birds, nor were our repeatability estimates any lower in our urban sample. Thus, while regional heterothermy during stress exposure may provide energetic benefits to some, but not all, individuals, enhanced use of this response to cope with urban pressures appears unlikely in our study species.

Keywords Urbanisation · Heterothermy · Stress · Thermoregulation · Repeatability

Introduction

Recent population estimates suggest that nearly 4.3 billion humans (or approximately 54% of the global population) now reside in urban environments, with urban living becoming normalised across the globe (reviewed in Lerch 2017). Clearly, such massive urban expansion cannot occur in a vacuum, and has already contributed to widespread reductions in habitat availability and quality for numerous species

(Xu et al. 2018; Freeman et al. 2019; discussed in Elmqvist et al. 2015). For this reason, understanding whether these species can adapt and persist within modern city-scapes has become an urgent priority among ecologists and conservationists today (e.g. Birnie-Gauvin et al. 2016; Ouyang et al. 2018).

For many vertebrates, a key problem with urban expansion is exposure to new challenges uncommon to rural environments, including noise, frequent human interaction, vehicle traffic, and in some cases, elevated depredation and inter- and intra-specific competition (Johnson et al. 2012; Hernández-Brito et al. 2014; Newsome et al. 2015; Vincze et al. 2017; reviewed in Lowry et al. 2013). Coping with these challenges is likely to require regular activation of energetically costly stress-physiological pathways (i.e. the hypothalamic–pituitary–adrenal axis: reviewed in Bonier 2012), which may place urban individuals at particular risk of physiological damage if energy is also required to be allocated elsewhere (Romero et al. 2009). Although recent meta-analyses provide little evidence for increased physiological

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